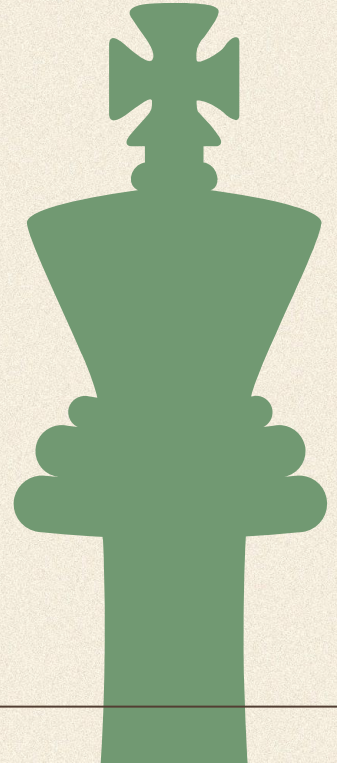


Project Joseph Hoane

Season Finale

IOWA STATE
UNIVERSITY





Introducing the Team

Brayden Hayworth

Connor Tynan

Evan Litzer

Jordan Rios



Project Overview

A Vision-Guided Chess Robot

- Fully autonomous robotic chess system
- Computer vision recognizes and maintains game state
- Utilizes chess engine to compute optimal move
- Motion planner translates move to robotic arm manipulation



Client Information

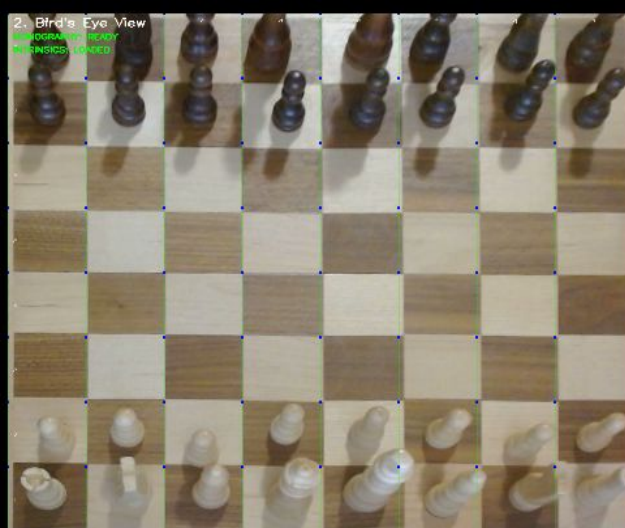
Bowen Weng

- Computer science assistant professor at Iowa State University.
- Researches robotics and autonomous systems.
- Oversees the Joseph Hoane project through all installments.



Vision Model From Fall / Background

- Tasked with robotic and game management focus.
 - Building upon fall semester vision module.
- Pipeline: **camera frames** → **FEN** (first part)
 - OpenCV → 1280x720 @ 30 FPS
 - Detects board using manual corner calibration and 9 methods. (fallback)
 - YOLOv8 model trained on all 12 piece types.
 - Pixel coordinate mapping that builds piece map → FEN string.
 - Calibration interface with keyboard controls.
- Problem: Retrain the piece detection model!



4. Analysis

Status: **SUCCESS**

FEN: 1N6/3B4/8/8/8/8/k6k/kbk5

Time: 0.064s



6. Controls

- q/ESC – Quit
- s – Manual capture
- a – Toggle auto-capture
- g – Stop game & save
- c – Toggle corner overlay
- p – Toggle piece overlay
- m – Cycle detection method
- o – Set method to 'auto'
- u – Use saved calibration
- r – Recalibrate
- [/] – YOLO conf – / +

Method: **saved_calibration**

YOLO conf: 0.25

Game Engine

- **Stockfish engine**
- Input FEN, output move (ex: e2e4).
- **Game manager** parses move, detects special cases, updates FEN, and sends move information to robot.
- Move sent to move planner through pipeline.
- Move planner receives move directive and plans a series of robot actions (move to xyz, grab, drop, etc.)
- Problems: Special moves, vision consistency, move and game state detection



piece/position/ranks w KQkq - 0 1



Vision-to-Engine Pipeline

1. Vision Model

- Input camera frames to initialize or maintain game state
- Outputs FEN string, piece map, board corners, and confidence level

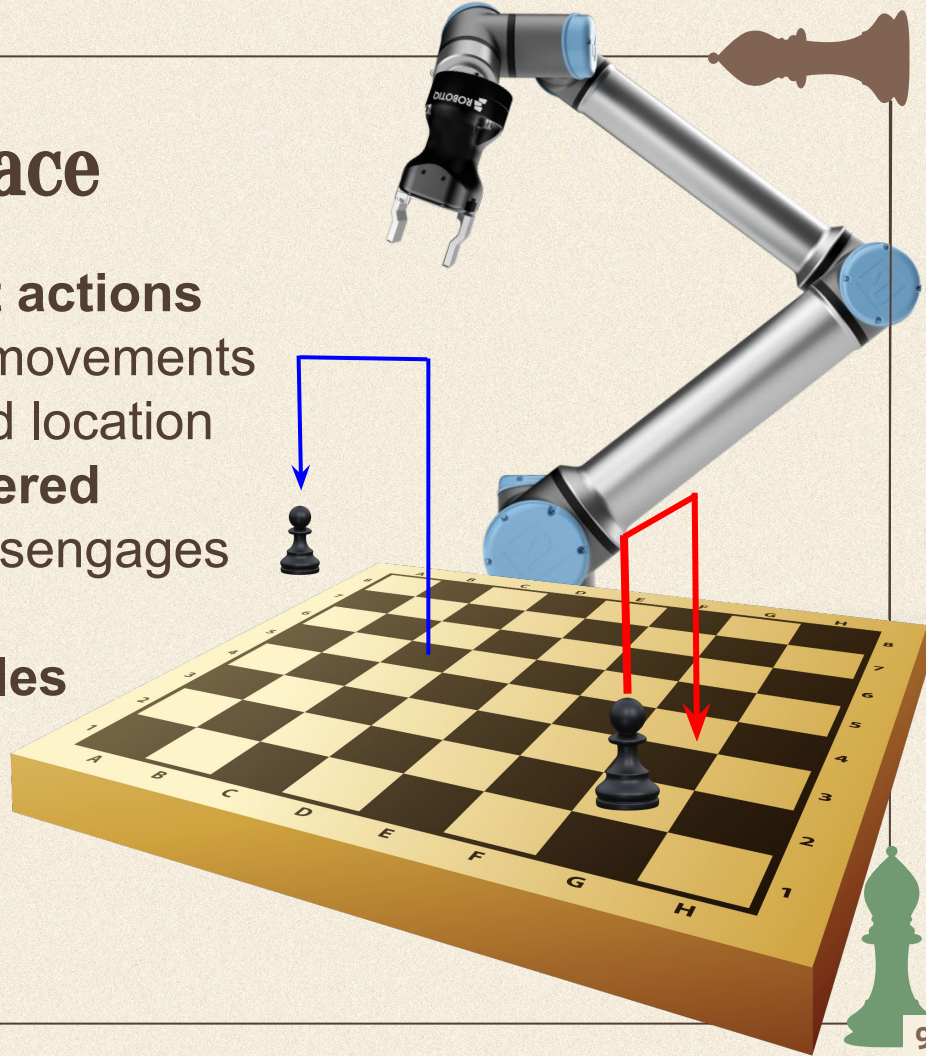
2. Game Engine / Chess Logic

- Takes FEN string as input
- Chess engine computes best response move
- Game state manager enforces chess-game rules, then sends move to motion planning
- Also detects special game states and is called on move detection



Designing Pick & Place

- **Goal: Two primitive movement actions**
 - On-board square to square movements
 - On-board square to off-board location
- **Subgoal: Ensure piece is centered**
 - Stabilize piece before arm disengages
- **Challenge: Various piece profiles**
 - Height: Pawn vs Queen
 - Shape: Knight vs Rook





Engine-to-Robotics Pipeline

3. Motion Planning & Control

- Task planner converts computed chess move into motions/tasks (pick up, move, place, capture, castle)
- Solve inverse kinematics (IK) to move to designated pose in space
- Send target torque control to each joint to carry out motions

4. Robotics Layer

- Take in target torque values for each joint
- Validate IK in MuJoCo simulation and execute arm manipulation
- Once in target position, send grasp/release signal to gripper



Virtual-to-Reality

- **Phase 1: Simulation**
 - MuJoCo simulation using ur10e model
 - Integrate vision-to-motion planning
- **Goal: Vision-in-the-loop Simulation**
 - Full pipeline demonstration in MuJoCo
- **Phase 2: Reality Check**
 - Full integration with ur10e arm
 - Adapt pipeline for ROS controllers

